Dave's Daily

AI • Bayesian optimization • US equities

tldr

1. Nvidia is in early talks to buy Reflection AI, to hire its staff and license the model, or to add to an $800 million stake. In April the chief executive named a $25 billion valuation before new money. No price and no deal are set.
2. Treasury wants the builders of models to pay when those models cause harm. The main US backing statute still requires intent. A Houston law professor says an agent could do what would be a crime for a person, and still leave nobody liable.
3. A new optimizer buys a cheap trial only when that trial would change beliefs about the expensive feasible tradeoff. On a five-input test with 100 observations, the next trial took a median of 103 seconds to choose. A deep-network method with the same goal took 3,354 seconds.

AI

The chip company wants a model company. On Saturday the Financial Times reported that Nvidia is in early talks to acquire Reflection AI or to deepen an investment, citing people with direct knowledge (Financial Times; Reuters). Nvidia has already put $800 million into the startup, the newspaper said. Reuters could not verify the report that day, and neither company commented then. On Sunday Bloomberg, citing one person familiar with the matter, said the talks could also be a supply of chips, that no terms are set, and that Reflection declined to comment (Straits Times).

What a deal looks like. The Financial Times lists a full purchase, an arrangement that hires the staff and licenses the technology, or a larger equity check. The hiring-and-license shape can skip a long merger review, and the talks can still end. Reflection was founded in 2024 by former DeepMind researchers Misha Laskin and Ioannis Antonoglou. In April Laskin told CNBC the company was raising at a $25 billion valuation before the new money. That figure is the April valuation. No purchase price has been named.

The model, and who pays. On 5 October Reflection released Beam, an open-weight model for coding and agents. The company says Beam costs less than many rivals and scores near GLM-5.2, an open model from China's Z.AI, on reasoning tests. Open weights mean the parameters are public. Bloomberg's Maggie Eastland, writing Friday evening, puts the legal question next to that industrial bet (Bloomberg). Treasury Secretary Scott Bessent told Congress last month, "The best way to guarantee safety is that the creators are liable for what they build and generate." Late Friday the White House Super Intelligence Force, chaired by intelligence chief Jay Clayton, said companies must disclose incidents and remedy the harm, and that delay "will not be tolerated." The Computer Fraud and Abuse Act requires prosecutors to prove intent. Gabriel Weil, of the University of Houston, told the New York City Council on Monday that agents could do what would be a tort or a crime for a human, "and yet no one could be liable under current law." Senators Josh Hawley and Chris Murphy plan a bill that would make developers liable for failing to limit a model's hacking. Representative Lori Trahan's draft, released Wednesday, would hold developers responsible even when they used reasonable care. Hypothesis: a statute that needs a human state of mind will not decide the next incident. The bills are an attempt to write a rule that can.

AI-related debt issuance

Billions of dollars, global

Gold is June, the peak month. September is the published round figure of about $23 billion. The axis starts at zero and stops at 150. Source: Financial Times, 10 October 2026, citing Morgan Stanley.

Markets

The borrowing paused. The Financial Times reported Saturday that new debt raised for AI fell sharply in September, citing Morgan Stanley (Financial Times). Global AI-related issuance was about $23 billion in September, against $113 billion in June, the year's peak. In the US investment-grade market, AI issuance was zero in September. Through September the year's total was $466 billion, against $216 billion for all of 2025. The figure shows those two months. Morgan Stanley calls September a pause: much of the borrowing was pulled into the first half, data-center permits and power are slowing projects, and higher yields are forcing some project loans to be rewritten. The bank expects high-grade cloud companies to return in the fourth quarter, at a slower pace than the first half, and puts 2027 capital spending by six large builders at about $1.4 trillion, above a published consensus near $1 trillion to $1.1 trillion. These notes describe the financing mix. They are not a forecast of share prices.

Bayesian optimization

Scaling: one score for the tradeoff, the constraint, and the price of the trial. Rikuto Matsumoto, Masanori Ishikura, and Masayuki Karasuyama (Nagoya Institute of Technology) choose both the design and the fidelity by how much a trial would teach about the feasible tradeoff at the expensive fidelity (arXiv:2610.10174, 7 Oct). A cheap trial counts when it changes the chance that the expensive output still lies inside that tradeoff. The chance is a Gaussian probability. Each objective and each constraint has its own multi-fidelity Gaussian process. A rival with the same goal, MF-JESMOC, fits a deep Gaussian process instead. On a problem with 5 inputs, 2 objectives, 2 constraints, and 100 observations, the median time to pick the next trial was 103.1 seconds for the new method and 3,354.1 seconds for the deep-process method. Five seeds, one thread of a Xeon Gold 5317. The quartiles were 102.5–121.6 and 3,309.6–3,364.0 seconds. Two faster methods that split the decision into separate rules took 12.7 and 13.2 seconds. The link to current model work is the same budget: a cheap measurement is worth buying only when it moves beliefs about the expensive measurement.

The experiment. Each run starts with five designs, measured at both fidelities, then has a budget of 50 expensive trials. On the synthetic tests a cheap trial costs 1 per output and an expensive trial costs 10. After ten cheap trials in a row, the next trial must be expensive. Ten seeds. The score is the hypervolume of designs recommended from the posterior mean, divided by the hypervolume of an approximate true feasible. The largest gap appears when the test functions are themselves draws from the Gaussian process, the case where the model matches the data. The same method also tunes LightGBM on the Covertype forest data: macro-F1 against a smaller model, with a floor on the worst-class recall, and 100 trees against 200. The candidate list is 512 fixed designs.

Data efficiency. There is no regret bound. The authors write that the error in the information approximation is still unknown. Hypothesis: the roughly 32-fold faster choice, relative to the deep-process method, comes from staying with an ordinary Gaussian process. Whether the cheaper trials buy a better tradeoff is in the figures, which the paper hoped does not reduce to one number.

Updated Sunday 11 October 2026. Same edition ships as phone and letter. The Financial Times reported the Nvidia talks and the debt slowdown on 10 October. Bloomberg's liability piece is dated 10 October. The optimizer paper is from 7 October. Market notes describe the financing mix, not a forecast. Created with mdvlm.